

# *Critical Ratio Technique*

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- 1. Helps determine the status of specific jobs**
- 2. Establishes relative priorities among jobs on a common basis**
- 3. Relates both stock and make-to-order jobs on a common basis**
- 4. Adjusts priorities automatically for changes in both demand and job progress**
- 5. Dynamically tracks job progress**

# Critical Ratio:

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- The critical ratio method assigns a priority that is a continually updated ratio between the time remaining until due date and the required job processing time. When used in conjunction with other jobs waiting to be processed, it is a relative measure of critical job order priority.

# Continue...

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- ❑ The critical ratio gives the highest priority to jobs that must be done to maintain a predetermined shipping schedule.
- ❑ Jobs that are falling behind a shipping schedule receive a ratio of less than 1...
- ❑ While a job receiving a critical ratio greater than 1 is ahead of schedule and is less critical.
- ❑ A job receiving a critical ratio score of 1.0 is precisely on schedule.

# *Critical Ratio (CR)*

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- ◆ An index number is found by dividing the time remaining until the due date by the work time remaining on the job
- ◆ Jobs with low critical ratios are scheduled ahead of jobs with higher critical ratios
- ◆ Performs well on average job lateness criteria

# Calculating Critical Ratio:

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- ❑ The critical ratio is calculated by dividing the remaining time until the date due by the remaining process time using the following formula:
- ❑  $\text{critical ratio} = \text{remaining time} / \text{remaining process time}$   
 $= (\text{Due Date} - \text{Today's date}) / \text{days of remaining process time}$

# Example:

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- ❑ On day 16, four jobs, A, B, C, and D, are on order for Ferguson's Kitchen Installation Service:
- ❑ Jobs on Order

| Job | Due Date | Days of Remaining Process Time |
|-----|----------|--------------------------------|
| A   | 27       | 8                              |
| B   | 34       | 16                             |
| C   | 29       | 15                             |
| D   | 30       | 12                             |

# Continue...

- Using this data, the critical ratios and priority order are computed.

| Critical Ratios and Priority |                   |          |
|------------------------------|-------------------|----------|
| Job                          | Critical Ratio    | Priority |
| A                            | $(27-16)/8=1.38$  | 4        |
| B                            | $(34-16)/16=1.13$ | 3        |
| C                            | $(29-16)/15=.87$  | 1        |
| D                            | $(30-16)/14=1$    | 2        |

- Job C has a critical ratio less than one indicating it has fallen behind schedule. Therefore, it gets the highest priority. Job D is exactly on schedule, but jobs B and A have respectively higher critical ratios indicating they have some slack time. This gives them respectively lower priorities.

Continue...

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# Critical Ratio

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Critical ratio:

- looks at time remaining between current time and due date
- considers processing time as a percentage of remaining time
- $CR = 1.0$  means just enough time
- $CR > 1.0$  more than enough time
- $CR < 1.0$  not enough time

# Example: Critical Ratio

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|            |             |            |              |
|------------|-------------|------------|--------------|
| □ T = 0    | Process     |            | Critical     |
| <u>Job</u> | <u>Time</u> | <u>Due</u> | <u>Ratio</u> |
| 1          | 6           | 18         | 3.0          |
| 2          | 2           | 6          | 3.0          |
| 3          | 3           | 9          | 3.0          |
| 4          | 4           | 11         | 2.75         |
| 5          | 5           | 8          | 1.6          |

Job 5 is done first.

# Example: Critical Ratio

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| □ T = 5    | Process     | Due -          | Critical     |
|------------|-------------|----------------|--------------|
| <u>Job</u> | <u>Time</u> | <u>Current</u> | <u>Ratio</u> |
| 1          | 6           | 13             | 2.17         |
| 2          | 2           | 1              | 0.5          |
| 3          | 3           | 4              | 1.33         |
| 4          | 4           | 6              | 1.5          |

Job 2 is done second.

# Example: Critical Ratio

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| □ T = 7    | Process     | Due -          | Critical     |
|------------|-------------|----------------|--------------|
| <u>Job</u> | <u>Time</u> | <u>Current</u> | <u>Ratio</u> |
| 1          | 6           | 11             | 1.84         |
| 3          | 3           | 2              | 0.67         |
| 4          | 4           | 4              | 1.0          |

Job 3 is done third.

# Example: Critical Ratio

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|            |             |                |              |
|------------|-------------|----------------|--------------|
| □ T = 10   | Process     | Due -          | Critical     |
| <u>Job</u> | <u>Time</u> | <u>Current</u> | <u>Ratio</u> |
| 1          | 6           | 8              | 1.84         |
| 4          | 4           | 1              | 0.25         |

Job 4 is done fourth, and job 1 is last.

# Critical Ratio Solution

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| Job   | Time | Done      | Due | Tardy    |
|-------|------|-----------|-----|----------|
| 5     | 5    | 5         | 8   | 0        |
| 2     | 2    | 7         | 6   | 1        |
| 3     | 3    | 10        | 9   | 1        |
| 4     | 4    | 14        | 11  | 3        |
| 1     | 6    | 20        | 18  | 2        |
| Total |      | <u>56</u> |     | <u>7</u> |

Mean flow time =  $56 / 5 = 11.2$

Average tardiness =  $7 / 5 = 1.4$

Number tardy = 4

Max Tardy = 3